Quantification: all 9 images
==========
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Columns 1 to 6 of 35

| image_label | lacuna count | interior lacuna count | median lacuna area (px^2) | roots per cell | ring length 30 px per cell (px) | ring length 60 px per cell (px) |
|---|---|---|---|---|---|---|
| 548_hyp_HP_z08c1-2 | 15 | 13 | 3031 | 10.7692 | 336.692 | 640.923 |
| 548_hyp_HP_z14c1-2 | 21 | 13 | 1979 | 8.7692 | 289.154 | 593.923 |
| 548_hyp_HP_z20c1-2 | 18 | 14 | 1453 | 9.6429 | 311.929 | 628.786 |
| 699_HYP_HP_z11c1-2 | 8 | 8 | 1244 | 8.75 | 337.125 | 647.875 |
| 699_HYP_HP_z20c1-2 | 11 | 11 | 1239 | 11 | 392.091 | 765.727 |
| 699_HYP_HP_z25c1-2 | 5 | 5 | 1302 | 5.4 | 258.6 | 601 |
| 701_z19c1-2 | 8 | 8 | 1667 | 10.375 | 431.125 | 1048.75 |
| 701_z26c1-2 | 10 | 10 | 1656.5 | 11.2 | 450.3 | 1023.1 |
| 701_z36c1-2 | 7 | 7 | 620.5 | 8.1429 | 409.429 | 1129.43 |

Columns 7 to 12 of 35

| image_label | field length density (px^-1) | perimeter per cell (px) | ring area 30 px per cell (px^2) | ring area 60 px per cell (px^2) | in-frame fraction 30 px per cell | in-frame fraction 60 px per cell |
|---|---|---|---|---|---|---|
| 548_hyp_HP_z08c1-2 | 0.025267 | 288.307 | 9624.92 | 21102.8 | 0.9809 | 0.9558 |
| 548_hyp_HP_z14c1-2 | 0.0277793 | 274.976 | 8774.31 | 19049.1 | 0.9755 | 0.9497 |
| 548_hyp_HP_z20c1-2 | 0.0255181 | 302.359 | 9380.07 | 20606.6 | 0.9836 | 0.9528 |
| 699_HYP_HP_z11c1-2 | 0.0217585 | 239.478 | 7955.75 | 20072.2 | 1 | 1 |
| 699_HYP_HP_z20c1-2 | 0.0177434 | 328.647 | 9852.09 | 22578.3 | 0.9784 | 0.9565 |
| 699_HYP_HP_z25c1-2 | 0.0167945 | 211.097 | 6763.4 | 17800 | 1 | 1 |
| 701_z19c1-2 | 0.0422046 | 311.228 | 9109.62 | 22309.4 | 1 | 0.9914 |
| 701_z26c1-2 | 0.0456065 | 337.156 | 9328.5 | 21741.6 | 1 | 0.9957 |
| 701_z36c1-2 | 0.048789 | 136.282 | 6094.14 | 17768.4 | 1 | 1 |

Columns 13 to 18 of 35

| image_label | ring density 30 px per cell (px^-1) | ring density 60 px per cell (px^-1) | roots per 100 px perimeter per cell | ring attached length 30 px per cell (px) | ring attached length 60 px per cell (px) | ring length weighted 30 px per cell (px) |
|---|---|---|---|---|---|---|
| 548_hyp_HP_z08c1-2 | 0.0347237 | 0.0302466 | 3.8147 | 307.308 | 472.538 | 384.866 |
| 548_hyp_HP_z14c1-2 | 0.032182 | 0.0311459 | 3.3092 | 259.769 | 460.154 | 330.52 |
| 548_hyp_HP_z20c1-2 | 0.0328728 | 0.0306568 | 3.4346 | 274.929 | 462.429 | 354.971 |
| 699_HYP_HP_z11c1-2 | 0.0423638 | 0.0319926 | 3.7474 | 299.375 | 501.625 | 379.269 |
| 699_HYP_HP_z20c1-2 | 0.0360491 | 0.0316939 | 3.2694 | 347.727 | 591.455 | 438.975 |
| 699_HYP_HP_z25c1-2 | 0.0357005 | 0.0321157 | 2.4402 | 211 | 431.6 | 290.297 |
| 701_z19c1-2 | 0.046898 | 0.0463718 | 3.8605 | 363.875 | 740.125 | 482.923 |
| 701_z26c1-2 | 0.0480684 | 0.0457018 | 3.5545 | 385.6 | 731.1 | 506.051 |
| 701_z36c1-2 | 0.0665947 | 0.0630826 | 6.4764 | 343.857 | 785.857 | 466.815 |

Columns 19 to 24 of 35

| image_label | ring length weighted 60 px per cell (px) | Sholl crossings 10 px per cell | Sholl crossings 20 px per cell | Sholl crossings 30 px per cell | field length density weighted (px^-1) | field length density without flagged regions (px^-1) |
|---|---|---|---|---|---|---|
| 548_hyp_HP_z08c1-2 | 731.929 | 8.8462 | 8.6154 | 7.7692 | 0.0288708 | 0.025267 |
| 548_hyp_HP_z14c1-2 | 677.208 | 6.9231 | 8.0769 | 8.1538 | 0.0316184 | 0.0277793 |
| 548_hyp_HP_z20c1-2 | 719.819 | 7.2857 | 7.7857 | 7.8571 | 0.0290688 | 0.024252 |
| 699_HYP_HP_z11c1-2 | 728.019 | 8.125 | 8.75 | 9.5 | 0.0243061 | 0.0206753 |
| 699_HYP_HP_z20c1-2 | 860.389 | 8.7273 | 8.9091 | 11 | 0.0198989 | 0.0167541 |
| 699_HYP_HP_z25c1-2 | 670.724 | 6.4 | 9.6 | 7 | 0.018758 | 0.0150603 |
| 701_z19c1-2 | 1178.63 | 9.875 | 12 | 12.125 | 0.0475535 | 0.0392254 |
| 701_z26c1-2 | 1145.97 | 10.5 | 11.8 | 13.2 | 0.0511931 | 0.0438318 |
| 701_z36c1-2 | 1276.87 | 8.1429 | 9.8571 | 13.8571 | 0.0547478 | 0.0548852 |

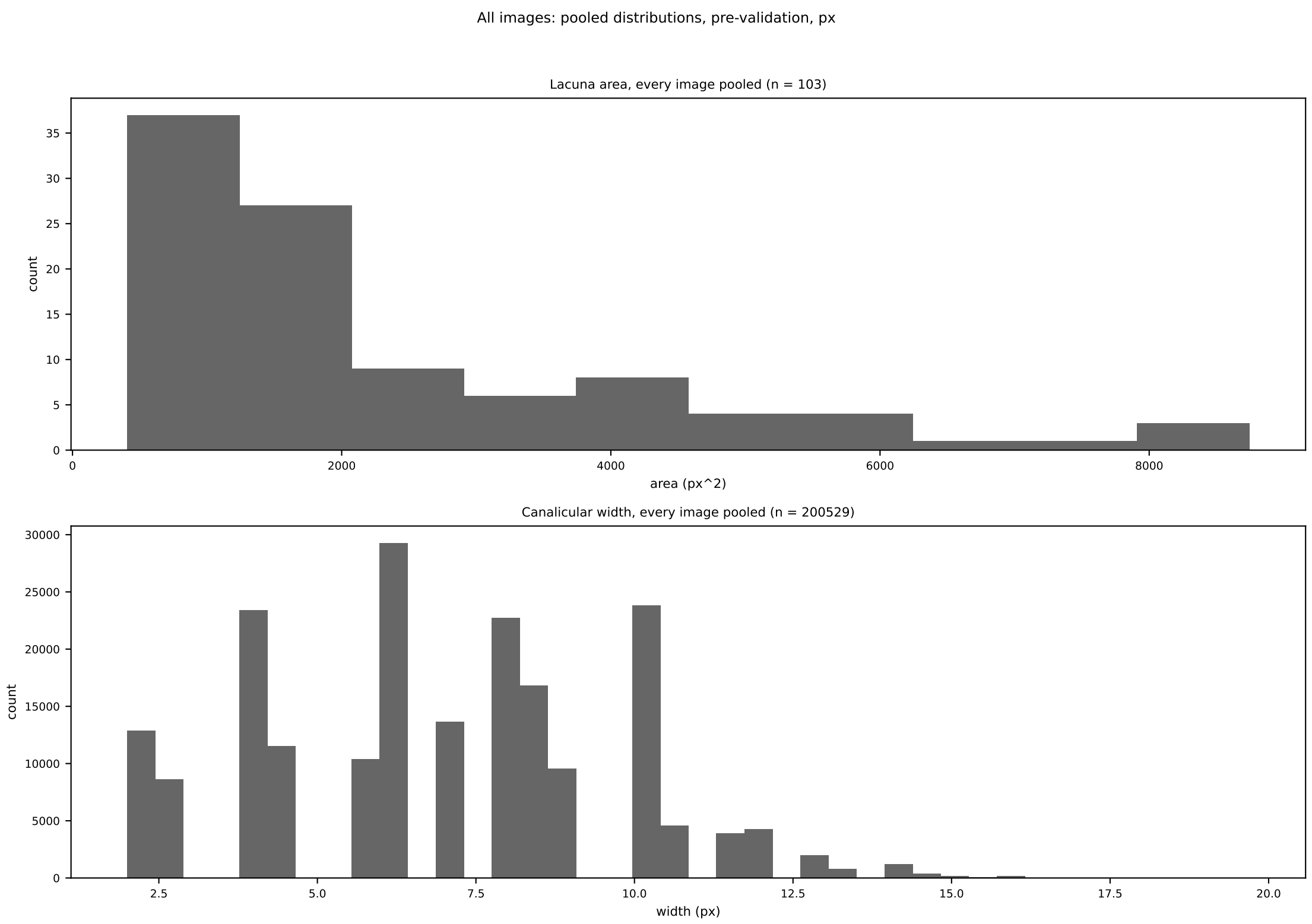
Columns 25 to 30 of 35

| image_label | field length density in ROI (px^-1) | loops per cell | share of a cell's edges on a loop | loops per field | share of field edges on a loop | canalicular width mean (px) |
|---|---|---|---|---|---|---|
| 548_hyp_HP_z08c1-2 | | 2.1538 | 0.1384 | 37 | 0.0956 | 7.6204 |
| 548_hyp_HP_z14c1-2 | | 3.3077 | 0.1959 | 74 | 0.1953 | 7.8593 |
| 548_hyp_HP_z20c1-2 | | 2.7143 | 0.1589 | 55 | 0.1388 | 7.4729 |
| 699_HYP_HP_z11c1-2 | | 2.125 | 0.1163 | 51 | 0.1681 | 7.315 |
| 699_HYP_HP_z20c1-2 | | 1.8182 | 0.099 | 24 | 0.0884 | 6.9355 |
| 699_HYP_HP_z25c1-2 | | 1.8 | 0.0625 | 30 | 0.1234 | 6.8823 |
| 701_z19c1-2 | | 10.75 | 0.2128 | 131 | 0.1986 | 6.5971 |
| 701_z26c1-2 | | 9.9 | 0.1718 | 162 | 0.1992 | 6.2156 |
| 701_z36c1-2 | | 27.4286 | 0.2935 | 245 | 0.244 | 6.0111 |

Columns 31 to 35 of 35

| image_label | canalicular width median (px) | canalicular width p10 (px) | canalicular width p90 (px) | canalicular width pixels used | canalicular width in ring 30 px per cell (px) |
|---|---|---|---|---|---|
| 548_hyp_HP_z08c1-2 | 8 | 4 | 10.7703 | 25179 | 7.3024 |
| 548_hyp_HP_z14c1-2 | 8 | 4 | 10.7703 | 27555 | 7.8403 |
| 548_hyp_HP_z20c1-2 | 8 | 4 | 10.7703 | 21895 | 7.259 |
| 699_HYP_HP_z11c1-2 | 8 | 4 | 10.198 | 16553 | 7.5953 |
| 699_HYP_HP_z20c1-2 | 7.2111 | 4 | 10 | 13941 | 7.3278 |

| | | | | | |
|---|---|---|---|---|---|
| 699_HYP_HP_z25c1-2 | 6.3246 | 4 | 10 | 12347 | 6.5562 |
| 701_z19c1-2 | 6.3246 | 2.8284 | 10 | 22002 | 6.5061 |
| 701_z26c1-2 | 6 | 2 | 10 | 27069 | 6.1446 |
| 701_z36c1-2 | 6 | 2 | 10 | 33988 | 6.3232 |


All images: pooled distributions, pre-validation, px
Lacuna area, every image pooled (n = 103)
count
area (px^2)
0
5
10
15
20
25
30
35
0
2000
4000
6000
8000
Canalicular width, every image pooled (n = 200529)
count
width (px)
0
5000
10000
15000
20000
25000
30000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0